DR. A. P. J. ABDUL KALAM TECHNICAL UNIVERSITY

UTTAR PRADESH, LUCKNOW



EVALUATION SCHEME

FOR

B.TECH. 3rd YEAR

Manufacturing Technology

[Effective from Session: 2024-25]

B. Tech.
Manufacturing Technology
Evaluation Scheme

SEMESTER-V													
Sl. No.	Code	Subject	Type	Category	Period			Evaluation Scheme			End Semester	Total	Credits
					L	T	P	CT	TA	Total	TE / PE		
1	BME 501	Heat & Mass Transfer	T	PC	3	1	0	20	10	30	70	100	4
2	BME 502	Machine Design	T	PC	3	1	0	20	10	30	70	100	4
3	BME 503	Industrial Engineering	T	PC	3	1	0	20	10	30	70	100	4
4		Departmental Elective-I	T	PE	3	0	0	20	10	30	70	100	3
5		Departmental Elective-II	T	PE	3	0	0	20	10	30	70	100	3
6	BME 551	Heat Transfer Lab	P	PC	0	0	2		50	50	50	100	1
7	BME 552	Machine Design Lab	P	PC	0	0	2		50	50	50	100	1
8	BME 553	Internet of Things Lab	P	PC	0	0	2		50	50	50	100	1
9	BME 554	Mini Project/Internship Assessment*	P		0	0	2		50	50	50	100	1
10	BNC501/ BNC502	Constitution of India/ Essence of Indian Traditional Knowledge	T	VA/HS	2	0	0	15	10	25	50		NC
11	MOOCs (Essential for Hons. Degree)												
	Total				**17**	**3**	**8**					**900**	**22**

*****The Mini Project or internship (4-5 weeks) conducted during summer break after IV semester and will be assessed during V semester.

B. Tech.
Manufacturing Technology
Evaluation Scheme

SEMESTER-VI													
Sl. No.	Code	Subject	Type	Category	Period			Evaluation Scheme			End Semester	Total	Credits
					L	T	P	CT	TA	Total	TE / PE		
1	BME 602	CAD/CAM	T	PC	3	1	0	20	10	30	70	100	4
2	BME 603	Theory of Machines	T	PC	3	1	0	20	10	30	70	100	4
3	BME 604	Tool Design	T	PC	3	1	0	20	10	30	70	100	4
4		Departmental Elective-III	T	PE	3	0	0	20	10	30	70	100	3
5		Open Elective-I	T	OE	3	0	0	20	10	30	70	100	3
6	BME 652	CAD/ CAM Lab	P	PC	0	0	2		50	50	50	100	1
7	BME 653	Theory of Machines Lab	P	PC	0	0	2		50	50	50	100	1
8	BME 654	Tool Design Lab	P	PC	0	0	2		50	50	50	100	1
9	BNC601/ BNC602	Constitution of India/ Essence of Indian Traditional Knowledge	T	VA/HS	2	0	0	15	10	25	50		NC
10	MOOCs (Essential for Hons. Degree)												
	Total				**17**	**3**	**6**					**800**	**21**

Departmental Electives

		Specialization-1	Specialization-2	Specialization-3
Semester	**Specialization**	**Manufacturing & Automation**	**Design Engineering**	**Industrial Management**
V	**Departmental Elective-I**	**BME 051**	**BME 053**	**BPI 051**
		Advance Manufacturing Processes	Finite Element Methods	Ergonomics & Workplace Design
	Departmental Elective-II	**BME 054**	**BME 056**	**BPI 052**
		Mechatronic Systems	Mechanical Vibrations	Facility Planning & Design
VI	**Departmental Elective-III**	**BME 061**	**BME 063**	**BPI 061**
		Industrial Robotics	Tribology	Decision Support & Intelligent System

BME 501 - Heat & Mass Transfer

The students will be able to		Blooms Taxonomy
CO1	Understand the fundamentals of heat and mass transfer.	K2
CO2	Apply the concept of steady and transient heat conduction.	K3
CO3	Apply the concept of thermal behavior of fins.	K3
CO4	Apply the concept of forced and free convection.	K3
CO5	Apply the concept of radiation for black and non-black bodies.	K3
CO6	Conduct thermal analysis of heat exchangers.	K4

Unit-1 (8-Hours)

Introduction to Heat Transfer

Introduction of thermodynamics and Heat Transfer, Modes of Heat Transfer: Conduction, convection and radiation, Effect of temperature on thermal conductivity of different types of materials, Introduction to combined heat transfer mechanism, General differential heat conduction equation in the rectangular, cylindrical and spherical coordinate systems, Initial and system boundary conditions.

Steady State one-dimensional Heat conduction

Simple and Composite Systems in rectangular, cylindrical and spherical coordinates with and without energy generation, Concept of thermal resistance, Analogy between heat and electricity flow, Thermal contact resistance and over-all heat transfer coefficient, Critical radius of insulation for cylindrical, and spherical bodies.

Unit-2 (6-Hours)

Fins

Heat transfer through extended surfaces and its classification, Fins of uniform cross-sectional area, Error in measurement of temperature of thermometer wells.

Transient Conduction

Transient heat conduction, Lumped capacitance method, Time constant, Unsteady state heat conduction in one dimension only, Heisler charts and their applications.

Unit-3 (10- Hours)

Forced Convection

Basic concepts: Hydrodynamic boundary layer, Thermal boundary layer, Approximate integral boundary layer analysis, Analogy between momentum and heat transfer in turbulent flow over a flat surface, Mixed boundary layer, Flow over a flat plate, Flow across a single cylinder and a sphere, Flow inside ducts, Thermal entrance region, Empirical heat transfer relations, Relation between fluid friction and heat transfer, Liquid metal heat transfer.

Natural Convection

Physical mechanism of natural convection, Buoyant force, Empirical heat transfer relations for natural convection over vertical planes and cylinders, horizontal plates, cylinders and sphere, combined free and forced convection, Effect of turbulence.

Unit-4 (8-Hours)

Thermal Radiation

Basic concepts of radiation, Radiation properties of surfaces, Black body radiation Planck's law, Wein's

displacement law, Stefan-Boltzmann law, Kirchhoff's law, Gray body, Shape factor, Black-body radiation, Radiation exchange between diffuse non-black bodies in an enclosure, Radiation shields, Radiation combined with conduction and convection; Absorption and emission in gaseous medium; Solar radiation; Greenhouse effect, Radiation network analysis.

Unit-5 (10-Hours)

Heat Exchanger

Different types of heat exchangers, Fouling factors, Overall heat transfer coefficient, Logarithmic mean temperature difference (LMTD) method, Effectiveness-number of transfer Unit (NTU) method and Compact Heat Exchangers.

Condensation and Boiling

Introduction of condensation phenomena, Heat transfer relations for laminar film condensation onvertical surfaces and on outside & inside of a horizontal tube, Effect of non-condensable gases, Drop wise condensation, Heat pipes, Boiling modes, pool boiling, Hysteresis in boiling curve, Forced convection boiling.

Introduction to Mass Transfer

Introduction of Fick's law of diffusion, Steady state equimolar counter diffusion, Steady state diffusion through a stagnant gas film, Heat and Mass Transfer Analogy -Convective Mass Transfer Correlations

Reference Books:-

1. Fundamentals of Heat and Mass Transfer, by Incroperra& DeWitt, John Wiley and Sons
2. Heat and Mass Transfer by Cengel, McGraw-Hill
3. Heat Transfer by J.P. Holman, McGraw-Hill
4. Heat and Mass Transfer by Rudramoorthy and Mayilsamy, Pearson Education
5. Heat Transfer by Ghoshdastidar, Oxford University Press
6. A text book on Heat Transfer, by Sukhatme, University Press.
7. Heat Transfer by Venkateshan, Ane Books Pvt Ltd
8. Schaum's outline of Heat Transfer by Pitts & Sisson McGraw-Hill
9. Heat and Mass Transfer by R Yadav, Central Publishing House

BME 502 - Machine Design

Course Outcomes: The student will be able to		Blooms Taxonomy
CO1	Recall the basic concepts of Solid Mechanics to understand the subject.	K2
CO2	Classify various machine elements based on their functions and applications.	K2
CO3	Apply the principles of solid mechanics to machine elements subjected to static and fluctuating loads.	K3
CO4	Analyze forces, bending moments, twisting moments and failure causes in various machine elements to be designed.	K4
CO5	Design the machine elements to meet the required specification.	K5

Unit-1 (8-Hours)

Introduction

Definition, Design requirements of machine elements, Design procedure, Standards in design, Standards designation of carbon & alloy steels, Selection of preferred sizes, Selection of materials for static and fatigue loads, Design against Static Load

Design against Fluctuating Loads

Cyclic stresses, Fatigue and endurance limit, Stress concentration factor, Stress concentration factor for various machine parts, Design for finite & infinite life, Soderberg, Goodman, Gerber criteria

Unit-2 (8-Hours)

Riveted Joints

Riveting methods, materials, Types of rivet heads, Types of riveted joints, Caulking and Fullering, Failure of riveted joint, Efficiency of riveted joint, Design of boiler joints, Eccentric loaded riveted joint

Welded Joints

Stress relieving of welded joints, Butt Joints, Fillet Joints, Strength of Butt Welds, Strength of parallel fillet welds, Strength of transverse fillet welds

Shafts

Cause of failure in shafts, Materials for shaft, Stresses in shafts, Design of shafts subjected to twisting moment, bending moment and combined twisting and bending moments, Shafts subjected to fatigue loads, Design for rigidity, Keys, Types of keys, Selection of square and flat keys, Strength of sunk key

Unit-3 (8- Hours)

Spur Gears

Tooth forms, System of gear teeth, contact ratio, Standard proportions of gear systems, Interference in involute gears, Backlash, Selection of gear materials, Gear manufacturing methods, Design considerations, Beam strength of gear tooth, Dynamic tooth load, Wear strength of gear tooth, Failure of gear tooth, Design of spur gears, AGMA and Indian standards.

Helical Gears

Terminology, Proportions for helical gears, Force components on a tooth of helical gear, Virtual number of teeth, Beam strength and wear strength of helical gears, Dynamic load on helical gears, Design of helical gears. Introduction, Classification and Applications of Bevel & Worm Gears

Unit-4 (8-Hours)

Sliding Contact Bearing

Types, Selection of bearing, Plain journal bearing, Hydrodynamic lubrication, Properties and materials, Lubricants and lubrication, Hydrodynamic journal bearing, Heat generation, Design of journal bearing.

Rolling Contact Bearing

Advantages and disadvantages, Types of ball bearing, Thrust ball bearing, Types of roller bearing, Selection of radial ball bearing, Bearing life, Selection of roller bearings, Dynamic equivalent load for roller contact bearing under constant and variable loading, Reliability of Bearing.

Unit-5 (8-Hours)

IC Engine Parts

Selection of type of IC engine, General design considerations, Design of Cylinder and cylinder head; Design of piston, piston ring and gudgeon pin;

Friction Clutches

Clutches, Difference between coupling and clutch, Single plate friction clutch, Torque transmitting capacity, Multi-Disk Clutches, Friction Material

Note: Design data book is allowed in the examination

Text Books:

1. Design of Machine Elements-V.B. Bhandari, McGraw Hill Co.
2. Design of Machine Elements, Sharma and Purohit, PHI.

Reference Books:

1. Mechanical Engineering Design, 9e – Joseph E. Shigely, McGraw Hill Education.
2. Machine Design-Maleev and Hartman, CBS Publishers.
3. Design of Machine Design-M.F. Spott, Pearson Education.
4. Elements of Machine Component Design, Juvinal & Marshek, John Wiley & Sons.
5. Machine design, Robert L. Norton, Pearson Education
6. Theory & Problem of Machine Design (Schaum's Outline Series) Hall, Holowenko, Laughlin, Tata McGraw Hill Co.
7. Machine Design-Sharma and Agrawal, S.K. Kataria & Sons.
8. Machine Design, U C Jindal, Pearson Education.

BME 503 - Industrial Engineering

Course Outcomes: The students will be able to		Blooms Taxonomy
CO1	Understand the concept of production system, productivity, facility and process planning in various industries	K2
CO2	Apply the various forecasting and project management techniques	K3
CO3	Apply the concept of break-even analysis, inventory control and resource utilization using queuing theory	K3
CO4	Apply principles of work study and ergonomics for design of work systems	K3
CO5	Formulate mathematical models for optimal solution of industrial problems using linear programming approach	K4

Unit-1 (9-Hours)

Overview of Industrial Engineering: Types of production systems, concept of productivity, productivity measurement in manufacturing and service organizations, operations strategies, liability and process design.
Facility location and layout: Factors affecting facility location; principle of plant layout design, types of plant layout; computer aided layout design techniques; assembly line balancing; materials handling principles, types of material handling systems, methods of process planning, steps in process selection, production equipment and tooling selection, group technology, and flexible manufacturing.

Unit-2 (8-Hours)

Production Planning and control: Forecasting techniques–causal and time series models, moving average, exponential smoothing, trend and seasonality; aggregate production planning; master production scheduling; materials requirement planning (MRP) and MRP-II; routing, scheduling and priority dispatching, concept of JIT manufacturing system.
Project Management: Project network analysis, CPM, PERT and Project crashing.

Unit-3 (8-Hours)

Engineering economy and Inventory control: Definition and Meaning of management, Methods of depreciation; break-even analysis, techniques for evaluation of capital investments, financial statements, time-cost trade-off, resource levelling; Inventory functions, costs, classifications, deterministic inventory models, perpetual and periodic inventory control systems, ABC analysis, and VED analysis.
Queuing Theory: Basis of Queuing theory, elements of queuing theory, Operating characteristics of a queuing system, Classification of Queuing models.

Unit-4 (7-Hours)

Work System Design: Taylor's scientific management, Gilbreths's contributions; work study: method study, micro-motion study, principles of motion economy; work measurement–time study, work sampling, standard data, Predetermined motion time system (PMTS); ergonomics; job evaluation, merit rating, incentive schemes, and wage administration.
Product Design and Development: Principles of product design, tolerance design; quality and cost considerations; product life cycle; standardization, simplification, diversification, value engineering and analysis, and concurrent engineering.

Unit-5 (8-Hours)

Operational Analysis: Formulation of LPP, Graphical solution of LPP, Simplex Method, Sensitivity Analysis, degeneracy and unbound solutions. Transportation and assignment models; Optimality test: the stepping stone method and MODI method, simulation.

Books and References:

1. Industrial Engineering and Production Management by Martand T Telsang S. Chand Publishing
2. Industrial Engineering and Production Management by M. Mahajan Dhanpat Rai & Co. (P) Limited
3. Industrial Engineering and Management by Ravi Shankar, Galgotia Publications Pvt Ltd
4. Production and Operations Management by Adam, B.E. & Ebert, R.J., PHI
5. Product Design and Manufacturing by Chitale A.V. and Gupta R.C., PHI
6. Operations Research Theory & Applications by J K Sharma, Macmillan India Ltd,
7. Production Systems Analysis and Control by J.L.Riggs, John Wiley & Sons
8. Automation, Production Systems & Computer Integrated Manufacturing by Groover, M.P. PHI
9. Operations Research, by A.M. Natarajan, P. Balasubramani, A. Tamilarasi, Pearson Education
10. Operations Research by P. K. Gupta and D. S. Hira, S. Chand & Co.

BME 551 - Heat Transfer Lab

The students will be able to		Blooms Taxonomy
CO1	Apply the concept of conductive heat transfer.	K3
CO2	Apply empirical correlations for both forced and free convection to determine the value of convection heat transfer coefficient	K3
CO3	Apply the concept of radiation heat transfer for black and grey body.	K3
CO4	Analyze the thermal behaviour of parallel or counter flow heat exchangers	K4
CO5	Conduct thermal analysis of a heat pipe	K4

List of Experiments

Minimum eight experiment of the following

1. To determine thermal conductivity of conductive material(s).
2. To determine thermal conductivity of insulating material(s).
3. To determine heat conduction through lagged pipe.
4. To determine heat transfer through fin under natural convection.
5. To determine the heat transfer Rate and Temperature Distribution for a Pin Fin.
6. Determination of thermal conductivity of different types of fluids.
7. Experiment on Stefan's Law - determination of emissivity, etc.
8. Experiment on convective heat transfer through flat plate solar collector.
9. To compare LMTD and Effectiveness of Parallel and Counter Flow Heat Exchangers.
10. To find the heat transfer coefficient for Forced Convection in a tube.
11. To find the heat transfer coefficient for Free Convection in a tube.
12. To conduct experiments on heat pipe.
13. To study the rates of heat transfer for different materials and geometries.
14. Visit to a Thermal Power Station for practical exposure.

BME 552 - Machine Design Lab

Course Outcomes: The student will be able to		Blooms Taxonomy
CO1	Apply the principles of solid mechanics to design various machine Elements subjected to static and fluctuating loads.	K3
CO2	Achieve an expertise in design of Sliding contact bearing in industrial applications.	K2
CO3	Write computer programs and validate it for the design of different machine elements	K4
CO4	Evaluate designed machine elements to check their safety.	K5

A Design of Machine Elements

1. Design a knuckle joint subjected to given tensile load.
2. Design a riveted joint subjected to given eccentric load.
3. Design of shaft subjected to combined constant twisting and bending loads
4. Design a transverse fillet welded joint subjected to given tensile load.
5. Design & select suitable Rolling Contact Bearing for a shaft with given specifications
6. Design a cylinder head of an IC Engine with prescribed parameters.
7. Design of Piston & its parts of an IC Engine

B. Computer Programs for conventional design Computer and Language

Students are required to learn the basics of computer language such as C/C++/MATLAB so that they should be able to write the computer program.

1. Design a pair of Spur Gear with given specifications to determine its various dimensions using Computer Program in C/C++.
2. Design a pair of Helical Gear with given specifications to determine its various dimensions using Computer Program in C/C++.
3. Design of Sliding Contact Bearing with given specifications & determine its various parameters using Computer Program in C/C++.

BME 553 - Internet of Things Lab

The students will be able to		Blooms Taxonomy
CO1	Understand Internet of Things and its hardware and software components	K2
CO2	Interface I/O devices, sensors & communication modules	K3
CO3	Remotely monitor data and control devices	K3
CO4	Design prototype of IoT based smart system	K4
CO5	Develop IoT based projects for real life problem	K6

List of Experiments:

S.No.	Name of Experiment	Outcome
1	Familiarization with concept of IoT, Arduino/Raspberry Pi and perform necessary software installation.	Will be able to understand IoT, Arduino/Raspberry Pi, and also able to install software setup of Arduino/ Respberry Pi
2	To interface motor using relay with Arduino/Raspberry Pi and write a program to turn ON/OFF motor.	Able to use relay to control motor and other mechanical devices
3	To interface sensors* with Arduino/Raspberry Pi and write a program to display sensors data on the computer screen.	Able to retrieve data from sensors and to display it on computer screen
4	To interface OLED with Arduino/Raspberry Pi and write a program to display sensor data on it.	Able to retrieve data from sensors and to display it on OLED
5	To interface sensor with Arduino/Raspberry Pi and write a program to turn ON/OFF Relay when sensor data is detected.	Able to control relay with help of microcontroller and sensors
6	To interface sensor with Arduino/Raspberry Pi and write a program to turn ON/OFF Solenoid valve when sensor data is detected.	Able to control Solenoid valve with help of microcontroller and sensors
7	To interface sensor with Arduino/Raspberry Pi and write a program to turn ON/OFF Linear Actuator when sensor data is detected.	Able to control linear actuator with help of microcontroller and sensors
8	To interface sensor with Arduino/Raspberry Pi and write a program to turn ON/OFF Starter Motor when sensor data is detected.	Able to control Starter Motor with help of microcontroller and sensors
9	To interface Bluetooth with Arduino/Raspberry Pi and write a program to send sensor data to smart phone using Bluetooth.	Able to communicate sensor data from microcontroller to smart phone

10	To interface Bluetooth with Arduino/Raspberry Pi and write a program to turn Actuators* ON/OFF when message is received from smart phone using Bluetooth.	Able to control actuators using mobile phone through Bluetooth
11	Write a program on Arduino/Raspberry Pi to upload Sensor data to things peak cloud.	Able to upload status of devices and sensors on web cloud
12	Write a program on Arduino/Raspberry Pi to retrieve sensors data from things peak cloud.	Able to retrieve status of devices and sensors from web cloud
13	Develop IoT based smart lock system for Motor cycle/Car	Able to develop smart lock system of motor cycle/car
14	Develop IoT based Smart water flow system	Able to develop smart water flow system
15.	Develop IoT based home security system	Able to develop smart home security system

Components required-

1. Arduino with cable
2. Raspberry Pi with cable and memory card
3. Node MCU
4. *Sensors-IR, LDR, DHT11 sensor, Push button, Pressure sensor, Temperature sensor, Vibration, Rotation, Location, Torque, Sound, Weight etc.
5. *Actuators-LED, Buzzer, Relay Switch, Motors, Motor Drivers, OLED, Display, Linear Actuator, Solenoid Valve, Starter Motor etc.
6. Bluetooth Module, Wi-fi Module, Ethernet Module
7. Smart Phone
8. Computer
9. Power Supply-5V, 12V, 3.3V
10. Internet facility

Semester – V: Departmental Elective – I
Specialization – Manufacturing & Automation

BME 051 - Advance Manufacturing Processes

CO	The students will be able to	Bloom Taxonomy
CO 1	Understand the physics of advanced machining process	K 2
CO 2	Analysis of advanced casting processes	K 2
CO 3	Selection of advanced welding process	K 2
CO 4	Analysis of advanced metal forming processes	K 2
CO 5	Product development using the advanced manufacturing processes	K 3

Unit-1 (8-Hours)
Advanced Machining Processes
Introduction, Process principle, Material removal mechanism, Parametric analysis and applications of processes such as Water jet machining (WJM), Abrasive water jet machining (AWJM), Electron beam machining (EBM), Laser beam machining (LBM) processes

Unit-2 (7-Hours)
Advanced Casting Processes
Metal mould casting, Continuous casting, Squeeze casting, Vacuum mould casting, Evaporative pattern casting, Ceramic shell casting

Unit-3 (8-Hours)
Advanced Welding Processes:
Details of electron beam welding (EBW), laser beam welding (LBW), ultrasonic welding (USW)

Unit-4 (8-Hours)
Advanced Metal Forming Processes
Details of high energy rate forming (HERF) process, Electro-magnetic forming, explosive forming Electro-hydraulic forming Stretch forming Contour roll forming

Unit-5 (7-Hours)
Additive Manufacturing Introduction:
Introduction to Prototyping, Traditional Manufacturing Vs. Additive Manufacturing, Need for time compression in product development, Usage of Additive Manufacturing parts
Powder Bed Fusion
Working Principal, Process Parameter, Advantage, Disadvantage and Application.

Reference Books:
1. "Materials and Processes in Manufacturing" (8th Edition), E. P. De Garmo, J. T Black, R. A. Kohser, Prentice Hall of India, New Delhi (ISBN 0-02-978760).
2. "Manufacturing Science" A. Ghosh, and A. K. Mallik, Affiliated East-West Press Pvt. Ltd. New Delhi.
3. "Nontraditional Manufacturing Processes", G.F. Benedict, Marcel Dekker, Inc. New York (ISBN 0- 8247-7352-7

Semester – V: Departmental Elective – I
Specialization – Design Engineering

BME 053 - Finite Element Methods

Course Outcome: Student will be able to		Bloom Taxonomy
CO1	Understand the basic concepts of FEM and its applications.	K2
CO2	Apply the procedure involved to solve a problem using Finite Element Methods.	K3
CO3	Develop the element stiffness matrices using different approach.	K3
CO4	Analyze 1D and 2D problem using different methods.	K4
CO5	Analyze the complex geometric problems through FEM software packages.	K4

Unit 1(8-Hours)
Introduction, exact solution vs approximate solution, principle of FEM, application of FEM, general procedure for finite element analysis, pre-processing, solution, post processing, Stresses and Equilibrium; Boundary Conditions.

Unit 2(8-Hours)
Strain-Displacement Relations, Stress–strain relations, Effect of temperature, various approximate methods: weighted residual method, variational or Rayleigh Ritz method, Galerkin's method, principle of minimum potential energy.

Unit 3(8-Hours)
Basic element shapes, generalized co-ordinates, polynomials, natural co-ordinates in one-two-and three-dimensions, Lagrange and Hermite polynomials, Application of Finite Element Methods to elasticity problems and heat conduction Problems.

Unit 4(8-Hours)
One dimensional problem of finite element model, Coordinates and Shape function, Potential-energy approach, Galerkine approach, Assembly of Global Stiffness Matrix and Load Vector. Plane trusses: Global and local coordinate system and stress calculation. Beams and Frames: finite element formulation and calculation of Shear Force and Bending Moment.

Unit 5(8-Hours)
Two-dimensional problem using Constant Strain Triangles and Four-node Quadrilateral, Problem modelling and Boundary conditions. Practical consideration in finite element applications, problem solving on a general purpose FEM software package like ANSYS, ABAQUS, NISA etc.

Text Books:

1. Chandrupatla, T. R. and Belegundu, A. K., Introduction to Finite Elements in Engineering, Pearson Education, India (2001).
2. Rao, S. S., Finite element method in engineering, 5th Edition, Pergaman Int. Library of Science, 2010.
3. Huebner, K. H., The Finite Element Method for Engineers, John Wiley, New York (2001).
4. Logan, D. L., A first course in the finite element method,6th Edition, Cengage Learning, 2016.

Semester – V: Departmental Elective – I

Specialization – Industrial Management

BPI 051 – Ergonomics & Workplace Design

Course Outcomes: The students will be able to		Bloom Taxonomy
CO1	Understand ergonomics, design principles for user-friendly systems, and factors influencing human performance and physical workload.	K 2
CO2	Understand the physiological principles, cognitive processing, mental workload, and factors influencing work-related musculoskeletal disorders.	K 2
CO3	Apply anthropometric principles to system design and understand human capabilities and limitations.	K 3
CO4	Evaluate occupational hazards & safety and job & environmental factors	K4
CO5	Apply ergonomics design principles to enhance performance, reduce fatigue, and improve human-computer interaction	K3
CO6	Design manufacturing workstations and tools.	K 5
CO7	Evaluate and measure sound, illumination, and occupational human vibration, and understand their effects on comfort, health, and performance in various tasks	K 4

Unit 1: Introduction (7-Hours)

Introduction to Ergonomics / Human Factors - overview and background; Design communication and ergonomics; Human-Machine system: Standard format for describing human-machine system, User friendly man machine environment system; Inter disciplinary nature of ergonomics; Modern ergonomics human performance – information processing – factors affecting human performance – physical workload and energy expenditure.

Unit 2: Human Physiology and Cognitive (8-Hours)

Physiological Principles: Sources of Energy, Nervous control of movements, structure of nervous system, Types of nervous system, Neurons, Action potential; Sodium potassium pump, innervations of muscles, Reflex-arc; Dynamics and static muscular work; Physiological (allowable limits and safety factors) and psycho sociological (behavior, cognitive issues, information processing and perception); Work-related Musculoskeletal Disorder, definition and Types of WMSDs, Factors affecting WMSDs; Cognitive processing such as perception, attention, learning, memory, and decision-making, information processing mental workload

Unit 3: Anthropometry and Workplace Design (8-Hours)

Introduction to Anthropometry and its application in design of system; Capabilities and limitations of people - physical (body structure, growth, anthropometry, biomechanics, and movement); Occupational hazards and safety, and environment factors affecting performance. Evaluation of facilities, environment, jobs and tasks, training methods and equipment, and user capabilities; Potential reduction of fatigue, errors, discomforts and unsafe acts; Ergonomics design principles and criteria, and checklist for ease and efficiency including Human-Computer-Interaction (HCI).

Unit 4: Workplace Design (8-Hours)

Manufacturing work-station design; determining work-station design parameters, Systematic approach for determining work-station design, determining work-station dimension. Tool evaluation and design: Principles of tool design (General principles, Anatomical concern, and Single handle); Attributes of common industrial hand tools, Attributes of common industrial power tools, Tool evaluation check list. Displays and controls, Ergonomic considerations in Visual Display Terminal (VDT) work station design and Office Design.

Unit V: Effect of Acoustic and Environmental Factors (8-Hours)

Sound and related studies: Definition, evaluation of noise, combining decibels. Levels and Spectra: Sound power level, sound intensity level, numerical problems on sound its measurement, Illumination and its measurement.

Occupational Human Vibration: Characteristics of vibration, wholebody and hand-arm vibration, Effect ofvibration on comfort, health and performance.

Studies related to the effect of Illumination, Heat and Indoor Air Quality on human performance of differenttypes of tasks.

Reference books:

1. R. S. Bridger, Introduction to Ergonomics, McGraw-Hill, Inc., 1995.
2. M. S. Sanders and E. J. McCormick, Human Factors in Engineering and Design, McGraw-Hill, Inc.,1993.
3. C. D. Wicknes, S. E. Gordon and Y. Liu, An Introduction to Human Factors Engineering, Longman, NY, 1997.
4. D. Chakrabarti, Indian Anthropometric Dimensions for ergonomic design practice, National Instituteof Design, Ahmedabad, 1997
5. J. Dul and B. Weerdmeester, Ergonomics for beginners a quick reference guide, Taylor & Francis, 1993.

Semester – V: Departmental Elective – II
Specialization – Manufacturing & Automation

BME 054 - Mechatronic Systems

Student will be able to		Bloom Taxonomy
CO1	Identify key elements of mechatronic and its representation by block diagram.	K2
CO2	Understand the concept of sensors and use of interfacing systems.	K2
CO3	Understand the concept and applications of different actuators	K2
CO4	Illustrate various applications of mechatronic systems.	K2
CO5	Develop PLC ladder programming and implementation in real life problem.	K5

Unit-1 (8-Hours)

Mechatronics & Its Scope

Fundamentals of Industrial automation**,** Introduction to Mechatronic Systems, Evolution, Scope, Application Areas, Basic Elements and Control of Mechatronic systems, Advantages and disadvantages of Mechatronics, Industrial applications of Mechatronics, autotronics, bionics, and avionics and their applications

Control System Concepts: Introduction to Control Systems, Elements of control system, Basic of open and closed loop control with example.

Unit-2 (8-Hours)

Sensor & Transducer

Definition and classification of sensor and transducer, performance terminology, static and dynamic characteristics, Principle of working and application of Inductive Proximity, Capacitive Proximity, Photoelectric, Ultrasonic, Magnetic, Hall Effect, Tactile Sensor, load cell, LVDT and interfacing sensors in Mechatronic system.

Unit-3 (8-Hours)

Actuation Systems

Fluid Based Actuation: Concept of Hydraulic and Pneumatic Actuation system, Oil and Air preparation Unit, Direction Control Valve, Pressure Control Valve, Single and doubly actuated systems, Actuators and Accumulators.

Electrical Actuation Systems: Introduction to Switching devices, Concept of Electro Mechanical Actuation, Solenoids and Solenoid Operated Direction Control Valves, Principle of working of DC and 3 Phase Induction Motor, Stepper motors and Servo Motors with their merits and demerits.

Unit-4 (8-Hours)

Industrial Controllers

Programmable Logic Controllers: Basic Structure, Types and Working Principle, Concept of Scan Cycle and Scan Time, IO's and its Types, Selection Criteria and Applications

Programming Techniques: Ladder diagram –Concept of Contacts and Coil, Latching/ Holding Circuit, Memory Bits, Timers and Counter.

Unit-5 (8-Hours)

Mechatronics Applications:

Control of conveyor motor, sorting and packaging Unit, pick and place robot, coin counter, operations of bottling plant, domestic washing machine, use of PLC for extending and retracting pneumatic pistons and their different combinations, automatic car park system, engine management system, other applications in manufacturing.

Text Books:

1. Rolf Isennann, " Mechatronics Systems", Springer, 2005.
2. W. Bolten, "Mechatronics", Pearson Education 2003.
3. HMT Ltd, "Mechatronics:, Tata McGraw Hill 1998.
4. K. P. Ramachandran, G.K. Vijayaraghavan, M.S. Balasundaram, Mechatronics - Integrated Mechanical Electronic Systems, Wiley.

Semester – V: Departmental Elective – II
Specialization – Design Engineering

BME 056 - Mechanical Vibrations

Student will be able to		Bloom Taxonomy
CO 1	Understand fundamentals of mechanical vibrations along with their classification.	K2
CO 2	Differentiate among single, two and multiple degree of freedom (DOF) systems.	K3
CO 3	Analyze, predict and measure the performance of systems undergoing single, two and multiple DOF.	K4
CO 4	Design systems with optimized vibration absorption capabilities.	K4
CO 5	Apply the fundamentals to the real life problems like whirling of shaft	K3
CO 6	Solve complicated mathematical models using Numerical methods and software applications.	K4

Unit-1 (10- Hours)

Introduction, Classification of Vibration Systems, Harmonic motion, Vector representation of harmonic motion, Natural frequency & response, Effects of vibration, superposition of simple harmonic motions, beats, Fourier analysis-analytical method.

Single Degree Freedom System, Equation of motion, Newton's method, D'Alembert's principle, Energy method etc., Free vibration, Natural frequency, Equivalent systems, Displacement, Velocity and acceleration, Response to an initial disturbance, Torsional vibrations, Damped vibrations, Vibrations of systems with viscous damping, Logarithmic decrement, Energy dissipation in viscous damping.

Unit –2 (8-Hours)

Single Degree Freedom: Forced Vibration Forced vibration, Harmonic excitation with viscous damping, steady state vibrations, Forced vibrations with rotating and reciprocating unbalance, Support excitation, Vibration isolation, Transmissibility, Vibration measuring instruments, Displacement, velocity, and acceleration measuring instruments

Unit- 3 (8-Hours)

Two Degree Freedom systems Introduction, Principal modes, Double pendulum, Torsional system withdamping, Coupled system, Principle of vibration absorber, Undamped dynamic vibration absorbers,Torsional vibration absorber, Centrifugal pendulum absorbers, Vibration isolators and Dampers.

Unit-4 (10- Hours)

Multi-degree Freedom system: Exact Analysis, Undamped free and forced vibrations of multi-degree freedom systems, influence coefficients, Reciprocal theorem, Torsional vibration of multi-degree rotor system, Vibration of gear system, Principal coordinates, Continuous systems- Longitudinal vibrations of bars, Torsional vibrations of circular shafts.

Multi Degree Freedom system: Numerical Analysis by Rayleigh's method, Dunkerley's, Holzer's and Stools methods, Rayleigh-Ritz method.

Unit- 5 (8-Hours)

Critical speed of shafts, Whirling of uniform shaft, Shaft with one disc with and without damping, Multi-disc shafts, Secondary critical speed.

Industrial case studies (any two) involving mechanical vibrations, their impact and performance analysis.

Introduction to the vibration analysis using MATLAB.

Books and References:

1. Mechanical Vibrations- V.P. Singh, Dhanpatrai & Co.
2. Mechanical Vibrations- G. K. Grover, Jain Brothers, Roorkee.
3. Mechanical Vibrations- Kelly
4. Mechanical Vibrations- Tse, Morse & Hinkle

Recommended software packages:

1. MATLAB
2. Any modelling and FEA tool like NX, Solid works etc.

Semester – V: Departmental Elective – II
Specialization – Industrial Management

BPI 052 – Facility Planning & Design

The students will be able to		Blooms Taxonomy
CO1	Understand the group technology, Plant Layout, material handling systems, & their Constituents.	K2
CO2	Understand the various plant layout design algorithms and quantitative models.	K2
CO3	Apply various graphical and computerized layout techniques & algorithms for plant layout development.	K3
CO4	Analyze various types of plant layouts, and factors affecting facility location and its layout.	K4
CO5	Able to suggest a material handling system for a plant under given constraints.	K4
CO6	Development of part families, machine cell formation for implementation of Group Technology and their balancing.	K4
CO7	Evaluate different plant layout alternatives within given constraints.	K5

Unit-1 (8-Hours)
Factory Planning: Introduction, factors to be considered
Plant Location and Site Selection: Objectives and importance of plant location and selection, Levels of plant location, rural, urban and suburban location of plants, factors influencing the plant location, optimum plant location, location theories.

Unit-2 (8-Hours)
Plant Layout: Introduction of production system, scope, objectives, importance, and types of plant layout, characteristics of a good plant layout, factors affecting plant layout, procedure of developing a plant layout, different graphical and computer based plant layout design techniques (e.g. ALDEP, CORELAP etc.), installation and evaluation of plant layout, optimum plant layout.

Unit-3 (8-Hours)
Group Technology: Definition, objectives, planning, part families and machine cell formation, product flow analysis, Rank order clustering, evaluation of machine cells, types of GT layout, benefits of GT, implementation of GT.

Unit-4 (8-Hours)
Line Balancing: introduction to assembly workstations, assembly line, Definitions, heuristic and analytical methods of balancing the assembly and production line, single and mixed model line balancing, alternatives to line balancing.

Unit-5 (8-Hours)

Materials Handling: Definition, scope, objectives, principles, importance, factors in materials handling problem, analysis of materials handling, types and selection of materials handling equipment's, aids and techniques in materials handling equipment selection. Planning of material flow, advantages of planned material flow, flow planning principles, flow patterns, analysis of material flow.

Recommended Books:

1. Francis, R.L., McGinnis, L.F., and White, J.A.,"Facility Layout and Location: An Analytical Approach", Prentice Hall of India 2004
2. Meyers, F.E., and Stephens, M.P., "Manufacturing Facilities Design and Material Handling", Prentice-Hall, Inc. 2000
3. Groover, M.P.,"Automation, Production Systems and Computer Integrated Manufacturing",2nd 2001 Ed., Pearson Education Inc. Delhi
4. Sule, D.R.,"Manufacturing Facilities-Location, Planning, and Design", PWS Publishing Company 1984
5. Tompkins, J.A., White, J.A., Bozer,Y.A.,Frazelle, E.H.,Tanchoco, J.M., and Tervino,J.,"Facilities Planning",2nd 1996 Ed., John Willey & Sons

BME 602 - CAD/CAM

Course Outcomes: The students will be able to		Blooms Taxonomy
CO1	Able to explain the 2D and 3D transformations, clipping algorithm, Manufacturing models and Metrics.	K2
CO2	Able to explain the fundamentals of geometric modeling, parametric curves, surfaces And Solids.	K2
CO3	Summarize the different types of Standard systems used in CAD.	K2
CO4	Able to apply NC & CNC programming concepts to develop part programme for Lathe & Milling Machines	K3
CO5	Understand the different types of techniques used in Cellular Manufacturing and FMS	K3

Unit-1 (8-Hours)

Introduction:

Product cycle- Design process- sequential and concurrent engineering- Computer aided design – CAD system architecture- Computer graphics – co-ordinate systems- 2D and 3D transformations homogeneous coordinates – Line drawing -Clipping- viewing transformation-Brief introduction to CAD and CAM – Manufacturing Planning, Manufacturing control- Introduction to CAD/CAM –CAD/CAM concepts —Types of production– Manufacturing models and Metrics– Mathematical models of Production Performance

Unit-2 (8-Hours)

Geometric modeling:

Representation of curves- Hermite curve- Bezier curve- B-spline curves-rational Curves-Techniques for surface modeling – surface patch-Coons and bi-cubic patches-Bezier and B-spline surfaces. Solid Modeling techniques-CSG and B-rep

Unit-3 (8-Hours)

CAD standards:

Standards for computer graphics- Graphical Kernel System (GKS) – standards for exchange images- Open Graphics Library (OpenGL) – Data exchange standards–IGES, STEP, CALS etc.–communication Standards

Unit-4 (8-Hours)

Fundamental of CNC and part programming.

Introduction to NC systems and CNC – Machine axis and Co-ordinate system- CNC machine tools- Principle of operation CNC- Construction features including structure-DrivesandCNCcontrollers-2D and 3D Machining on CNC-Introduction of Part Programming, types–Detailed Manual part programming on Lathe & Milling machines using G code sand M codes- Cutting Cycles, Loops, Sub program and Macros- Introduction of CAM package.

Unit-5 (8- Hours)

Cellular manufacturing and flexible manufacturing system (FMS)

Group Technology (GT), Part Families–Parts Classification and coding–Simple Problems in Opitz Part Coding system–Production flow Analysis–Cellular Manufacturing–Composite part concept–Types of Flexibility–FMS–FMS Components – FMS Application & Benefits – FMS Planning and Control– Quantitative analysis in FM

Reference Books:

1. V. Rajaraman. Computer Oriented Numerical Methods, PHI Learning, 4th Edition, 2019.
2. E Zimmer M Groover. CAD/CAM Computer Aided Design and Manufacturing, Pearson, 2014.
3. Na Vitalii Ivanov, Yiming Rong, Justyna Trojanowska, Joachim Venus. Advances in Design, Simulation and Manufacturing, Springer, 1st Edition, 2019.
4. Tiago Franco, Beatriz Costa, Maria Grilo. Product Design Process, Imaginary Cloud Limited, 2019.

BME 603 - Theory of Machines

Course Outcomes: The students will be able to		Blooms Taxonomy
CO1	Understand the principles of kinematics and dynamics of machines.	K2
CO2	Calculate the velocity and acceleration for 4-bar and slider crank mechanism	K3
CO3	Develop cam profile for followers executing various types of motions	K3
CO4	Apply the concept of gear, gear train and flywheel for power transmission	K3
CO5	Apply dynamic force analysis for slider crank mechanism and balance rotating &reciprocating masses in machines.	K3
CO6	Apply the concepts of gyroscope, governors in fluctuation of load and brake & dynamometer in power transmission	K3

Unit-1 (09-Hours)

Introduction, mechanisms and machines, kinematics and kinetics, types of links, kinematic pairs and their classification, types of constraint, degrees of freedom of planar mechanism, Grubler's equation, mechanisms, inversion of four bar chain, slider crank chain and double slider crank chain.

Velocity analysis: Introduction, velocity of point in mechanism, relative velocity method, velocities in four bar mechanism, instantaneous center.

Acceleration analysis: Introduction, acceleration of a point on a link, acceleration diagram, Corioli's component of acceleration, crank and slotted lever mechanism.

Unit-2 (10-Hours)

Cams: Introduction, classification of cams and followers, cam profiles for knife edge, roller and flat faced followers for uniform velocity, uniform acceleration

Gears and gear trains: Introduction, classification of gears, law of gearing, tooth forms and their comparisons, systems of gear teeth, length of path of contact, contact ratio, minimum number of teeth on gear and pinion to avoid interference, simple, compound, reverted and planetary gear trains, sun and planet gear train.

Unit-3 (08-Hours)

Force analysis: Static force analysis of mechanisms, D'Alembert's Principle, dynamics of rigid link in plane motion, dynamic force analysis of planar mechanisms, piston force and crank effort. Turning moment on crankshaft due to force on piston, Turning moment diagrams for single cylinder double acting steam engine, four stroke IC engine and multi-cylinder engines, Fluctuation of speed, Flywheel.

Unit-4 (09-Hours)

Balancing: Introduction, static balance, dynamic balance, balancing of rotating masses, two plane balancing, graphical and analytical methods, balancing of reciprocating masses, balancing of single cylinder engine.

Governors: Introduction, types of governors, characteristics of centrifugal governors, gravity controlled and spring controlled centrifugal governors, hunting of centrifugal governors, inertia governors. Effort and Power of governor

Unit-5 (09-Hours)

Brakes and dynamometers: Introduction, Law of friction and types of lubrication, types of brakes, effect of braking on rear and front wheels of a four wheeler, dynamometers, belt transmission dynamometer, torsion dynamometer, hydraulic dynamometer

Gyroscope: Space motion of rigid bodies, angular momentum, gyroscopic couples, gyroscopic stabilization, ship stabilization, stability of four wheel and two wheel vehicles moving on curved paths.

Text / Reference Books

1. Kinematics and dynamics of machinery: Wilson and Sadler, Third edition, Pearson.
2. Theory of Mechanisms and Machines: Amitabh Ghosh and Ashok Kumar Mallik, Third Edition Affiliated East-West Press.
3. Theory of Machines and Mechanisms: Joseph Edward Shigley and John Joseph Uicker, Jr. Oxford University Press
4. Kinematics and dynamics of machinery: R L Norton, McGraw Hill
5. Theory of Machines: S.S. Rattan, McGraw Hill
6. Theory of Machines: Thomas Bevan, CBS Publishers.

Suggested Software

Mech Analyzer

BME 604 – Tool Design

Course Outcomes: The students will be able to		Blooms Taxonomy
CO1	Understand the design criterion for single and multipoint cutting tools for different machines.	K2
CO2	Design jigs and fixtures with required tolerances and economy.	K3
CO3	Design the dies for casting and injection moulding processes.	K3
CO4	Design the dies for casting and injection moulding processes.	K3
CO5	Select a suitable control system, drives and kinematic mechanism for a machine tool.	K3

Unit-1 Cutting Tools (8-Hours)

Single point Cutting Tools: Requirements of a tool designer, general tool design procedure, Definition and geometry of single point cutting tool, design of single point lathe tool: design of shank dimension using strength and rigidity considerations for rectangular, square and round cross section and selection of tool geometry; solid type tool, brazed tip tool, long index able insert, throwaway index able insert types and chip breakers.

Multi Point Cutting Tool: Geometry of drill tool, end and face mill cutter; Drill bit design of elements like back taper, web thickness, land width, margin, flute length and cross section and selection of tool geometry; design of milling cutter: design of elements like number of teeth and height circular pitch, body thickness, chamfer width, fillet radius and selection of tool geometry.

Unit-2 Jigs and Fixtures (8-Hours)

Jigs & Fixtures: Tooling, die-punch, mould, work holding and fixturing; Introduction to tooling design; procedure of designing; role of Tolerance and clearance; economic considerations and calculations; principle of location of the work piece, degrees of freedom, references surfaces, resting components, fixture elements for surface concentric and radial locations; clamping of the work-piece; review of cutting forces, principles and methods of clamping; quick clamping devices, standards; guiding elements for tools, gauge for cutter, jig bushes, indexing methods, design of jigs/fixtures/accessories for drilling, milling, turning, broaching, grinding, and welding.

Unit-3 Sheet Metal, Bending and Drawing (8-Hours)

Design of dies: Simple piercing and blanking process, design of inverted die, compound die, progressive dies, rules for developing stock, strip layouts for progressive dies, types of progressive dies, load centre, analytical and graphical method to determine load centre (i.e. centre of pressure), miscellaneous dies, shaving, horn, cam actuated and precision lamination dies, fine blanking dies, principles - design considerations.

Bending dies: Theory of bending, blank development, spring back effect, spring back factor, methods of correction to overcome spring back, types of bending dies, pressure pad dies, forces in bending, construction and working principles, press brake tooling, curling, flanging principles of stretch forming – stretch forming dies.

Unit-4 Die Casting Dies and Injection Moulding (8-Hours)

Design of dies for metal casting : Casting dies, terminology applicable to process, terminology applicableto dies, alignment of metal flow in hot chamber, horizontal cold chamber and vertical cold chamber machines, modification for casting of deep core or with limitations of stroke, design for

various elements, effect of off-centre cavity layout, necessity of balancing, types of dies, parameters influencing the runner and gate design, ejection mechanism, ejection elements and various locations, die locking mechanism, trimming , types of trim dies and alloys suitable for die making.

Injection Moulding: Injection moulding machine and its elements, general configuration of a mould, 2- plate and 3-plate mould, gate, runner, parting surface, ejection system, core and cooling system; introduction to compression die, transfer die, blow moulding dies, extrusion dies, forming and calendaring dies.

Unit 5: Control, drives and structure (8-Hours)

Introduction to kinematics of machine tools, motion transmission, design of hydrostatic, hydrodynamic and antifriction guideways, design of spindles, design of speed gear boxes and feed box, feed drives, stepped and stepless regulations of speed, feed diagram, ray diagrams and design considerations.

Vibration in machine tools, lubrication, rigidity, and reliability in machine tools

Text Book

1. Machine Tool Design and Numerical Control by N.K. Mehta, McGraw Hill

Reference Books:

2. Principles of Machine Tools by G.C. Sen and A. Bhattacharya, Central Book Agency
3. Design of Machine Tools by D. K Pal and S. K. Basu, Allied Publishers
4. Machine Tool Design, Vol. I, II, III and IV, N. S. Acherkhan, Mir Publishers
5. Jigs and Fixtures, by Joshi, Tata McGraw Hill, New Delhi.
6. Theory and Application of Metal Cutting, by Juneja, Wiley Eastern Ltd
7. Basic Die Making & Advance Die Making by D. Eugene Ostergaard, McGraw Hill
8. Injection Mould Design by R.G.WPye, Longman Scientific & Technical Publishers

BME 652 - CAD/CAM Lab

Total Eight Experiments are to carried out. FOUR Experiments each from CAD and CAM.

A. CAD Experiments

1. Line Drawing or Circle Drawing experiment: Writing and validation of computer program.
2. Geometric Transformation algorithm experiment for translation/rotation/scaling: Writing and validation of computer program.
3. Design of machine component or other system experiment: Writing and validation of computer program.
4. Understanding and use of any 3-D Modeling Software commands.
5. Pro/E/Idea etc. Experiment: Solid modeling of a machine component
6. Writing a small program for FEM for 2 spring system and validation of program or using a FEM Package
7. Root findings or curve fitting experiment: Writing and validation of computer program.
8. Numerical differentiation or numerical integration experiment: Writing and validation of computer program.

B. CAM Experiments

1. To study the characteristic features of CNC machine
2. Part Programming (in word address format) experiment for turning operation (including operations such as grooving and threading) and running on CNC machine
3. Part Programming (in word address format or ATP) experiment for drilling operation (point to point) and running on CNC machine
4. Part Programming (in word address format or ATP) experiment for milling operation (contouring) and running on CNC machine
5. Experiment on Robot and programs
6. Experiment on Transfer line/Material handling
7. Experiment on difference between ordinary and NC machine, study or retrofitting
8. Experiment on study of system devices such as motors and feed back devices
9. Experiment on Mechatronics and controls

BME 653 - Theory of Machines Lab

The students will be able to:		Blooms Taxonomy
CO1	Demonstrate various mechanisms, their inversions and brake and clutches in automobiles	K2
CO2	Apply cam-follower mechanism to get desired motion of follower.	K3
CO3	Apply the concepts of gears and gear train to get desired velocity ratio for power transmission.	K3
CO4	Apply the concept of governors to control the fuel supply in engine.	K3
CO5	Determine the balancing load in static and dynamic balancing problem	K3

List of Experiments

(Minimum eight experiments out of the following)

NOTE: Student has to write computer program in C / C++ / Python and to run to compute the output values for at least ONE experiments.

1. To study various types of kinematics links, pairs, chains & Mechanisms
2. To study Whitworth Quick Return Motion Mechanisms, Reciprocating Engine Mechanism, and Oscillating Engine Mechanism
3. To study of inversions of four bar linkage
4. To study of inversions of single/double slider crank mechanisms
5. To study various types of gear (Helical, cross helical, worm, bevel gear) and gear profile (involute and cycloidal) and condition for interference Helical, cross helical, worm, bevel gear
6. To compute the output velocity invarious gear trains
7. To study gyroscopic effects through models
8. To determine gyroscopic couple on Motorized Gyroscope
9. To perform experiment on dead weight type governor to prepare performance characteristic Curves, and to find stability & sensitivity
10. To perform experiment on spring controlled governor to prepare performance characteristic Curves, and to find stability & sensitivity
11. To determine whirling speed of shaft theoretically and experimentally
12. To perform the experiment for static / dynamic balancing
13. To perform experiment on brake
14. To perform experiment on clutch
15. To perform the experiment for static / dynamic balancing.
16. To perform experiment on longitudinal vibration
17. To perform experiment on transverse vibration

BME 654- Tool Design Lab

The students will be able to		Blooms Taxonomy
CO1	Analyze the cutting forces in orthogonal machining.	K4
CO2	Develop a single point cutting tool using tool grinder	K6
CO3	Develop jig and fixtures for various applications	K6
CO4	Develop mould for injection moulding/ metal forming	K6
CO5	Develop die for balking/drawing/bending	K6

List of Experiments: (Minimum eight experiments are to be performed out of the following)

1. Design and prototype development of angle plate jig.
2. Design and prototype development of milling fixture.
3. Design and prototype development of turning fixture.
4. Design and prototype development of two plate mould.
5. Design of different types of gates used in injection moulding.
6. Design of Two plate moulds for metal forming work.
7. Design of Blanking Die with fixed stripper for a given blank.
8. Design of drawing die for a cylindrical/ rectangular component.
9. Design of 'V' Bending Die for 90-degree V shape bend.
10. Design of 'U' Bending Die for U shape bend.
11. Design and manufacturing of a single point cutting tool using tool grinder.
12. Measurement and analysis of cutting forces in orthogonal turning for different materials at different process parameters.

Semester – VI: Departmental Elective – III

Specialization – Manufacturing and Automation

BME 061 - Industrial Robotics

The students will be able to:		**Blooms Taxonomy**
CO1	Explain robot anatomy, classification, and applications of robots	K2
CO2	Understand the various sensors, grippers and its selection in robotics	K2
CO3	Obtain basic idea on working principle of various drives, actuators and control concepts	K2
CO4	Program different robot operations and appreciate applications of robots in industry.	K3
CO5	Understand the safety and basics of Artificial Intelligence for recent updates in robotics	K2

Unit-1 (8-Hours)

Introduction to robotics: Brief History, Basic Concepts of Robotics such as Definition, Three laws, Elements of Robotic Systems i.e. Robot anatomy, DOF, Misunderstood devices etc., Classification of Robotic systems on the basis of various parameters such as work volume, type of drive, etc., Associated parameters i.e. resolution, accuracy, repeatability, dexterity, compliance, RCC device etc., Introduction to Principles & Strategies of Automation, Types & Levels of Automations, Need of automation, Industrial applications of robot.

Unit-2 (8-Hours)

Grippers and Sensors for Robotics: Grippers for Robotics - Types of Grippers, Guidelines for design for robotic gripper, Force analysis for various basic gripper system. Sensors for Robots - Types of Sensors used in Robotics, Classification and applications of sensors, Characteristics of sensing devices, Selections of sensors. Need for sensors and vision system in the working and control of a robot.

Unit-3 (8-Hours)

Drives and Control for Robotics: Drive - Types of Drives, Types of transmission systems, Actuators and its selection while designing a robot system. **Control Systems**: Types of Controllers, Introduction to closed loop control.

Unit-4 (8-Hours)

Programming and Languages for Robotics: Robot Programming: Methods of robot programming, WAIT, SIGNAL and DELAY commands, subroutines, Programming Languages: Generations of Robotic Languages, Introduction to various types such as VAL, RAIL, AML, Python, ROS etc., Development of languages since WAVE till ROS.

Unit-5 (8-Hours)

Related Topics in Robotics: Socio-Economic aspect of robotisation. Economical aspects for robot design, Safety for robot and standards, Introduction to Artificial Intelligence, AI techniques, Need and application of AI, mobile robotics, New trends & recent updates in robotics. Mobile

Robot locomotion: Types of locomotion, hopping robots, legged robots, wheel edrobots, stability, manoeuvr ability, controllability

Text Books/References:

1. S. K. Saha, Introduction to Robotics 2e,TATA McGraw Hills Education(2014)
2. Asitava Ghoshal, Robotics: Fundamental concepts and analysis, Oxford University Press (2006)
3. Dilip Kumar Pratihar, Fundamentals of Robotics, Narosa Publishing House,(2019)
4. R.K. Mittal, I. J. Nagrath, Robotics and Control, TATA McGraw Hill Publishing Co Ltd, New Delhi (2003)
5. S. B. Niku, Introduction to Robotics –Analysis, Contro, Applications, 3rd edition, John Wiley & Sons Ltd., (2020)
6. R. Siegwart, I. R. Nourbakhsh, "Introduction to Autonomous Mobile Robots", The MIT Press, 2011

Semester – VI: Departmental Elective – III

Specialization – Design Engineering

BME 063 - Tribology

Student will be able to		Bloom Taxonomy
CO 1	Identify and explain various frictions and wear mechanisms.	K2
CO 2	Select proper lubricants for different applications.	K3
CO 3	Select suitable lubrication methods in different bearings.	K3
CO 4	Study the surfaces coating techniques for reduction of wear.	K3
CO 5	Analyze the impact of friction in various kinematic pairs.	K4

Unit–1 (8-Hours)

Lubrication and Lubricants

Introduction to tribology, tribology in industry, basics modes of lubrication, oil viscosity, temperature and pressure dependence of viscosity, Viscosity index, viscosity measurement, properties of lubricants, temperature characteristics of lubricants, lubricant impurities and contaminants, mineral oils based lubricants, synthetic oils based lubricants, emulsions and aqueous lubricants, greases, and lubricant additives.

Unit–2 (8-Hours)

Friction and Wear

Friction-causes of friction, theories of dry friction; adhesion theory, abrasive theory, junction growth theory, laws of rolling friction, friction measurement, friction instabilities.

Wear- classification; abrasive wear, erosive wear, cavitation wear, adhesive wear, corrosive wear, oxidative wear, fatigue wear, factors affecting wear, measurement of wear, theories of wear, approaches to friction control and wear prevention.

Unit–3 (8-Hours)

Lubrication of Bearings

Theory of hydrodynamic lubrication, mechanism of pressure development in oil film, jet lubrication, mist lubrication, lubrication utilizing under race passage, concept of journal bearing, minimum oil film thickness, porous bearings, flat plate thrust bearing, tilting pad bearings, hydrostatic lubrication, squeeze film lubrication, elasto-hydrodynamic lubrication, rolling element bearings, gas lubricated bearings, and hybrid bearings.

Unit–4 (8-Hours)

Solid Lubrication and Surface Treatment

Lubrication by solids, friction and wear characteristics of lamellar solids, reduction of friction by soft metallic films, deposition methods of solid lubricants, techniques for producing wear resistant coatings, characteristics of wear resistant coatings.

Unit–5 (8-Hours)

Friction, Lubrication and Wear in Kinematic pairs

The concept of friction angle, friction stability, friction in slideways, friction in screws with square threads, friction in screws with triangular threads, mechanism and operation of plate clutch, cone clutch, rim clutch, centrifugal clutch, and belt drives, tribo design aspects of labyrinth seals, analysis of line contact

lubrication, analysis of point contact lubrication, cam follower system, traction in the contact zone, and hysteresis losses.

Books and References:

1. Fundamentals of Engineering Tribology with Applications by Harish Hirani, Cambridge English (2017)
2. Applied Tribology (Bearing Design and Lubrication), by Michael M Khonsari, John Wiley & Sons(2001).
3. Principles of Tribology, by J Halling, The Macmillan Press Ltd, London, (1975).
4. Friction, Wear, Lubrication:A textbook in Tribology, by Ludema K C, CRC Press, (2010).
5. Fundamentals of Machine Elements, B.J. Hamrock, B.O. Jacobson & S.R. Schmid, McGraw-Hill Inc.,(1998).
6. Fundamentals of Mechanical Component Design, by K.S. Edwards & R.B. McKee, McGraw-Hill Inc.,(1991).
7. Mechanical Engineering Design by J.E. Shigley and C R Mischke, Tata McGraw-Hill Publishing Company Limited, (2003).
8. Tribophysics, by N.P. Suh Prentice-Hall, (1986).
9. Friction, Wear, Lubrication: A Textbook in Tribology, by Kenneth C Ludema, LayoAjayi, CRC Press(2019).

Semester – VI: Departmental Elective – III
Specialization – Industrial Management

BPI 061 – Decision Support & Intelligent System

Course Outcomes: The students will be able to:		**Blooms Taxonomy**
CO1	Understand the fundamentals of decision support systems, computerized decision aids, expert systems and executive information systems.	K2
CO2	Understand the concept of hybrid support system for decision making	K2
CO3	Understand different components of DSS like Data warehousing, Data mining, modeling and analysis.	K2
CO4	Understand the basics of Artificial intelligence and expert systems.	K2
CO5	Examine the uses of various mathematical models, heuristics and simulation as a subsystem of DSS.	K3

UNIT-1: (8-Hours)
Management Support Systems: An Overview: Managers and Decision-Making; Managerial Decision- Making and Information Systems; Managers and Computer Support; Computerized Decision Support and the Supporting Technologies.

A Framework for Decision Support; The Concept of Decision Support Systems; Group Support Systems; Enterprise Information Systems; Knowledge Management Systems; Expert Systems; Advanced IntelligentDecision Support Systems; Hybrid Support Systems.

Decision-Making Systems, Modelling, and Support: Decision-Making: Introduction and Definitions; Systems; Models; Phases of the Decision-Making Process.

UNIT-2: (8-Hours)
Decision Support Systems: Different types of Managerial Decision Problems and the Role of a DSS in solving them, DSS Configurations; Characteristics and Capabilities of DSS; Components of DSS; theData Management Subsystem.

Modelling and Analysis: Modelling; Classification of Models, Purpose of Modelling, Solution Techniques: Optimization, Heuristics, and Simulation, Desirable features for Models in DSS, Models and Managers Static and Dynamic Models; Certainty, Uncertainty, and Risk; Influence Diagrams; MSS Modelling with Spreadsheets; Decision Analysis of a Few Alternatives.

Decision Support System Development: Introduction to DSS Development; The Traditional System Development Life cycle; Alternative Development Methodologies.

UNIT-3: (8-Hours)
Knowledge Management
Introduction to Knowledge Management; Organizational Learning and Transformation; Knowledge Management Initiatives; Approaches to Knowledge Management; Information Technology in KnowledgeManagement.

Knowledge Management Systems Implementation; Roles of People in Knowledge Management; Ensuring Success of Knowledge Management.

UNIT-4: (8-Hours)

Artificial Intelligence and Expert Systems: Representation in logic and schemas, semantic networks, production rules and frames, inference techniques – DSS applications.

Social networking: package choices- knowledge security, integrating with web -based and internal operational & support systems, change management, reward systems continuous improvement, Intellectual Property Rights.

UNIT-5: (8-Hours)

Advanced Intelligent Systems: Machine-Learning Techniques; Case-Based Reasoning; Basic Concept of Neural Computing; Learning in Artificial Neural Networks; Genetic Algorithms Fundamentals; DevelopingGenetic Algorithm Applications; Fuzzy Logic Fundamentals.

Intelligent Systems over the Internet: Web-Based Intelligent Systems; Intelligent Agents: An Overview; Characteristics of Agents; need of Intelligent Agents; Classification and Types of Agents; Internet-Based Software Agents; DSS Agents and Multi-Agents.

Books and References:

1. Decision Support Systems and Intelligent Systems, Seventh Edition, Efraim Turban, Jay E.Aronson,Richard V. McCarthy, Prentice-Hall of India, 2007.
2. Decision Support Systems, A Knowledge-Based Approach, Clyde W. Holsapple andAndrew B.Whinston.
3. Business Intelligence and Analytics: Systems for Decision Support" by Ramesh ShardaandDursun Delen.
4. Decision Support and Business Intelligence Systems" by Efraim Turban and Ramesh E Sharda.
5. Decision Making and Knowledge Decision Support Systems" by Anna Maria Gil-LafuenteandConstantin Zopounidis.
6. Decision Support, Analytics, and Business Intelligence (Information Systems Collection)" byDaniel JPower.
7. Fuzzy Sets, Decision Making, and Expert Systems (International Series in ManagementScienceOperations Research)" by Hans-Jürgen Zimmermann.
8. Decision Support Systems for Business Intelligence by Vicki L. Sauter